CREDIT CARD FRAUD DETECTION PROJECT

Introduction

Credit Card Fraud Detection is a Machine Learning project used to identify fraudulent transactions from genuine transactions. It is a classification problem where the output is either fraud or non-fraud. Fraud detection systems help banks and financial institutions reduce financial losses and improve customer security. Machine Learning algorithms analyze transaction patterns to detect suspicious activities. This project demonstrates how data-driven techniques can improve fraud prevention. Different algorithms are applied and compared for better accuracy and performance.

Dataset Description

The dataset used is Creditcardfraud.csv, which contains transaction-related information. Each row represents a credit card transaction. The dataset contains numerical features generated through preprocessing techniques. The target variable is **Class**, where 0 represents a normal transaction and 1 represents a fraudulent transaction. The dataset is highly imbalanced, making fraud detection more challenging and realistic.

Input and Output Variables

Input Features	Description
Time	Time is used as a transaction-related feature for fraud prediction.
V1	V1 is used as a transaction-related feature for fraud prediction.
V2	V2 is used as a transaction-related feature for fraud prediction.
V3	V3 is used as a transaction-related feature for fraud prediction.
V4	V4 is used as a transaction-related feature for fraud prediction.
V5	V5 is used as a transaction-related feature for fraud prediction.
V6	V6 is used as a transaction-related feature for fraud prediction.
V7	V7 is used as a transaction-related feature for fraud prediction.
V8	V8 is used as a transaction-related feature for fraud prediction.
V9	V9 is used as a transaction-related feature for fraud prediction.

Output Variable

Class — The predicted transaction category: 0 = Genuine Transaction 1 = Fraudulent Transaction

Data Preprocessing

Data preprocessing is an important step in Machine Learning. The dataset is loaded using Pandas. Features and target variables are separated. The dataset is split into training and testing sets. Since fraud datasets are imbalanced, preprocessing helps improve prediction quality. Preprocessing removes inconsistencies and prepares data for model training.

Feature Scaling

Feature scaling standardizes feature values. It improves model performance and convergence speed. Algorithms like Logistic Regression work better when numerical values are scaled properly.

Machine Learning Algorithms Used

1. Logistic Regression

Logistic Regression is a classification algorithm used for binary prediction problems. It predicts whether a transaction is fraudulent or genuine based on probability. It is simple, efficient, and commonly used as a baseline model.

2. Random Forest Classifier

Random Forest is an ensemble learning algorithm that uses multiple decision trees. It improves accuracy and reduces overfitting. It works well for fraud detection tasks and handles large datasets efficiently.

Model Evaluation

Models are evaluated using Accuracy Score and Classification Report. The classification report includes Precision, Recall, and F1-score. Higher accuracy and balanced precision-recall values indicate better fraud detection performance.

Results

Different algorithms produce different results. Logistic Regression provides a simple and fast baseline model, while Random Forest generally provides higher accuracy and better fraud detection capability. Ensemble methods often perform better in handling complex fraud patterns.

Conclusion

This project demonstrates how Machine Learning can detect fraudulent credit card transactions. Different algorithms were applied and compared to improve prediction performance. Data preprocessing and feature handling play an important role in achieving better accuracy. This project provides practical experience in classification problems and fraud analytics.

What I Learned

Data preprocessing techniques Handling imbalanced datasets Working with classification algorithms Model training and testing Evaluation using accuracy and classification metrics Comparing machine learning models Fraud detection concepts